SafetyScan:
A PPE Detection System for Workplace Safety

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ABSTRACT
Ensuring the proper use of personal protective equipment (PPE) was essential for maintaining safety in construction and industrial workplaces. This project proposed the development of an AI-based system that automatically detects PPE, such as hard hats, safety vests, and masks, in worksite images using computer vision techniques. The core of the system utilized a convolutional neural network-based object detection model, such as YOLO, trained on publicly available PPE datasets. The model identified and localized PPE items in images, and its performance was evaluated using metrics such as mean Average Precision (mAP) and Precision-Recall curves. In addition to the vision component, the system integrated natural language processing to generate descriptive outputs and reinforcement learning to optimize system decision policies. Ethical considerations, including privacy and responsible surveillance, were also examined to ensure responsible use of the technology.

Keywords
Personal Protective Equipment (PPE) Detection, Object Detection, Computer Vision, Deep Learning, Convolutional Neural Networks (CNN), YOLO, Workplace Safety, Reinforcement Learning, Natural Language Processing

1. INTRODUCTION
Maintaining adherence to Personal Protective Equipment (PPE) policies is important for workforce safety and overall worksite safety. However, this practice is often performed manually, which can be inefficient, inconsistent, and difficult to sustain for continuous monitoring. With the growing use of deep learning in computer vision, object detection models such as YOLO have made automated PPE monitoring in images a possible approach. Even so, the performance of these models still needs to be evaluated before they can be considered for practical use.

1.1 Problem Statement, Success Metrics, and Constraints

This project examined the use of computer vision for detecting personal protective equipment (PPE) in workplace images, particularly in construction and industrial contexts. PPE compliance is often monitored manually, and this project explored whether an automated image-based approach could be used to identify items such as hard hats, safety vests, and masks. The motivation of the study was to investigate a possible method for supporting workplace safety monitoring through object detection.

The system's performance was discussed using common object detection metrics, including mAP@0.5, precision, recall, and inference latency. These metrics were relevant because they reflected both detection performance and speed. Several constraints were also relevant to the project, including the limited scale and diversity of the available dataset, limited computational resources for training and experimentation, the difficulty of detecting PPE in complex workplace scenes, and ethical concerns related to privacy and surveillance.

2. REVIEW OF RELATED LITERATURE

2.1 Transition to Deep Learning for PPE Monitoring

The use of Artificial Intelligence (AI) in construction safety monitoring became increasingly popular in recent years, particularly for personal protective equipment (PPE) detection [1], [10]. Traditional safety monitoring in construction sites often relied on manual inspections or hardware-based sensors, which were costly, inconsistent, and difficult to scale [2]. Because of these limitations, researchers increasingly shifted toward computer vision and deep learning techniques for automated PPE compliance monitoring [2], [7]. Previous vision-based studies demonstrated that deep learning could already handle complex construction environments better than traditional methods [3], [10]. In particular, the Faster Region-based Convolutional Neural Network (R-CNN) architecture was used to detect workers not wearing hard hats in far-field surveillance videos, showing that deep learning models could identify safety violations even in wide-background scenes and across different worker postures [3]. However, while two-stage detectors such as Faster R-CNN offered high detection accuracy, they were often computationally expensive and less suitable for real-time applications [7]. As a result, single-stage detectors such as You Only Look Once (YOLO), which offered a more practical balance between speed

and accuracy for on-site deployment, became more popular [2], [10], [7].

2.2 Performance of Single-Stage Detectors

Among modern object detection methods, the YOLO family became one of the most widely used frameworks for real-time PPE detection because of its speed and accuracy. Earlier studies showed that YOLOv3 could detect workers and identify PPE such as hard hats and safety vests in real time [10], while later work using transfer learning improved its performance and achieved an F1-score of 0.96 [2]. This metric was particularly valuable in machine learning because it provided the harmonic mean of precision and recall, balancing the importance of both finding all safety violations and minimizing false alarms [4]. More recent comparisons of YOLOv3, YOLOv4, and YOLOv7 reported up to 97% mean Average Precision (mAP) at around 25 Frames Per Second (FPS), proving that the YOLO framework was both accurate and efficient enough for real-world compliance monitoring [7].

2.3 Recent Advances in YOLO-Based PPE Detection

Recent YOLO developments continued to push performance boundaries. A YOLOv5-based web platform for monitoring hard hats and vests achieved a 91.5 mAP, proving these models were ready for accessible, user-facing applications [9]. Benchmarking YOLOv8 on the Construction-Helmet-Vest (CHV) and SHEL5K datasets also showed significant gains in precision and recall, confirming that YOLOv5 and YOLOv8 were ideal for lightweight, efficient PPE systems [1].

2.4 Overcoming Environmental Constraints

Detecting PPE remained difficult in complex worksites with cluttered backgrounds, poor lighting, and small objects viewed from a distance [6], [8]. To address this, SD-YOLOv5s introduced a DilateFormer attention mechanism to better spot small targets like helmets, reaching an Average Precision (AP) of 93.7% [6]. Similarly, the Efficient Sub-Pixel Convolutional Network-YOLO framework combined YOLOv8 with a super-resolution neural network to recover fine details in low-light conditions, achieving an mAP@0.5 of 0.922 [8].

2.5 Synthesis and Relevance to the Present Study

Overall, the reviewed literature showed that deep learning, specifically YOLO-based single-stage detectors, was the most effective approach for real-time PPE monitoring in construction. While these models offered high speed and accuracy, research identified persistent challenges such as small object detection, variable lighting, and the need for contextual verification of equipment. This study addressed these gaps, evaluated modern YOLO iterations, such as YOLOv8, and explored additional logic to verify PPE was actually being used correctly, rather than just detected.

3. MODEL & METHODS

3.1 Modeling Architecture and System Components

The proposed system adhered to an architecture blending computer vision, processing of data, and decision-making. The use of YOLOv8 for real-time object detection of PPE such as hardhats, vests, and masks. Including a training pipeline, data processing of images with related objects ensures PPE compliance through evaluation, and transfer learning to improve on accuracy. Lastly, a Natural Language Processing module is used for generating simplified results textually, in accordance with performance metrics like precision, latency, recall, and mAP.

3.2 Training Details and Reproducibility

The model was trained using Ultralytics YOLOv8 for 50 epochs on the Hard hat workers dataset from Roboflow. Training was performed on a CPU-based system using Python 3.12.10 and Ultralytics 8.4.30.

The trained model contained 73 layers, 3,006,038 parameters, and 8.1 GFLOPs. Validation results showed an mAP@0.5 of 0.958 and an mAP@0.5:0.95 of 0.679. To support reproducibility, the project kept a record of the dataset, software versions, and model settings used during training.

3.3 Ethics, Policy, Intended Use, and Limitations

In compliance with data protection principles, the system utilizes a publicly accessible dataset. Avoiding facial recognition, the system is limited to PPE, ensuring privacy and data processing of personal information. Finally, the use of the object detection system is open through either webcam or images. System limitations include the risks of detection quality due to poor lighting, occlusions against PPE, and smaller-in-view objects, in line with limited or unbalanced training data potentially causing data bias, and real-time performance being hardware-dependent.

4. DATA & RESULTS

4.1 Dataset Source, License, and Collection Notes

The dataset used in this project was the “Hard Hat Workers Computer Vision Model” available on Roboflow and uploaded by Northeastern University - China, and maintained by Joseph Nelson. It was an object detection dataset containing approximately 7,035 images and three labeled classes: head, helmet, and person. These annotations made it suitable for training YOLO-based PPE detection models, particularly for detecting workers and protective headgear in workplace scenes.

The images showed workers in construction and industrial environments, making the dataset relevant for PPE monitoring tasks. According to the dataset page, the license was CC0 1.0 Universal (Public Domain), which allows reuse with attribution. In this project, the dataset was used for academic and research purposes.

4.2 Preprocessing and Data Splits

The dataset was acquired from Roboflow and converted into a YOLOv8 format for compatibility. Where dataset cleaning, in improving model performance and involved in data preprocessing, included the removal of the “person” class to reduce noise, leaving the head and helmet class for focus, helping in helmet compliance detection. In addition, image normalization and consistency checks were implemented in ensuring the quality of the data. Subsequently, data splitting was conducted into training and validation sets, where it enables the model to learn patterns while being evaluated via performance metrics.

4.3 Evaluation Metrics and Baselines

System performance was evaluated using common object detection metrics such as precision, mAP, latency, and recall. These metrics were used to examine the detection accuracy and efficiency of the model in identifying PPE-related classes. Baseline evaluation considered the model's initial detection performance before any further adjustments or refinements. Where applicable, comparisons were made to determine whether changes in model configuration or class selection affected overall detection results.

4.4 Ablation Studies

Ablation analysis in this project was limited, but one meaningful experimental change involved removing the "person" class from the dataset during preprocessing. This adjustment was intended to reduce unnecessary class overlap and focus the model more directly on helmet-related detection. The comparison was relevant for examining whether a simplified class structure could improve task focus and reduce confusion during prediction.

4.5 Error and Slice Analysis

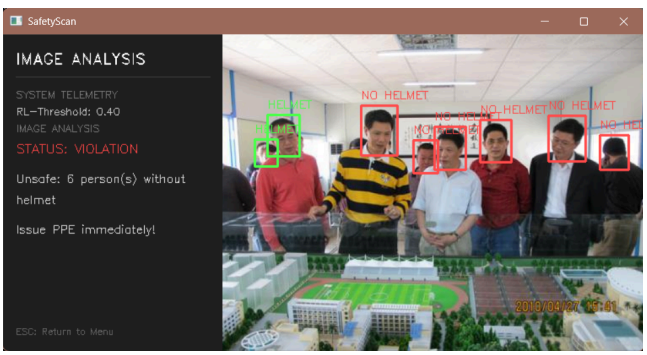

Figure 1. *SafetyScan screenshot showing an incorrect missing hard hat alert.*

Error analysis considered cases in which the system could generate incorrect violation alerts even when PPE items appeared to be present. In one example, the interface reported "MISSING: Hardhat" despite workers appearing to wear hard hats in the displayed scene. Detection boxes in the same example were labeled as "Safety Vest" in regions that did not clearly match the vest alone, which may indicate class confusion or incorrect localization. Such cases were relevant for examining possible weaknesses in detection and alert logic.

Slice analysis was also relevant for examining model behavior across different viewing conditions, such as screen-displayed images, crowded scenes, and images containing multiple workers. These cases helped identify practical limitations that may require further improvement.

5. MODEL CARD SUMMARY

The model used in this project was a YOLOv8-based object detection system for PPE-related monitoring in workplace images. It was mainly used to support helmet compliance detection by identifying the head and helmet classes in construction and industrial scenes. The model was trained on the "Hard Hat Workers Computer Vision Model" from Roboflow Universe, which contained about 7,035 images and was licensed under CC0 1.0 Universal (Public Domain).

Training was performed for 50 epochs using Ultralytics YOLOv8 with Ultralytics version 8.4.30 and Python 3.12.10 on a CPU-based system with an 11th Gen Intel Core i7-1165G7 processor. Validation results showed an mAP@0.5 of 0.958 and an mAP@0.5:0.95 of 0.679. Although the model showed strong detection performance, it could still be affected by class confusion, incorrect alerts, lighting changes, occlusion, complex backgrounds, and hardware limitations.

The model was intended only for academic, experimental, and decision-support use. It was not designed for facial recognition, identity tracking, or automatic disciplinary action. Any real-world use of the system should include human oversight, clear user notice, limited storage of image data, and regular checking of model performance in the target environment.

6. HOW TO RUN

The SafetyScan PPE Detection system is designed to ensure reproducibility and efficiency are consistent across all implementations, which warrant swift execution using standard Python environments. Its setup comprises Python version and essential dependencies installation, virtual environment creation, as well as a model retraining step-by-step guide on Google Colab. Utilizing an online GPU source to maintain training time requirement compliance (≤90 min. completion).

6.1 System Requirements

To run the program, the following requirements must first be met to ensure a properly equipped and error-free environment:

- Python versions 3.10.xx or higher.
- Required libraries: ultralytics, opencv-python, torch, torchvision, numpy, matplotlib, pillow and tk.
- Operating System: Compatible with both Windows and Linux (the system itself if tested on Windows).
- Recommended Device Specifications: Optional GPU, with CPU-supported execution available.

6.2 Installation

The following are the step-by-step installation guide (on bash terminal or local device or, if using Visual Studio Code, the project-root new terminal):

1. Clone the repository.

```
git clone <repository-link>
cd SafetyScan
```

Alternative: Go to GitHub repository and download the project as ZIP.

2. Create a virtual environment (optional).

```
python -m venv venv
venv\Scripts\activate   # Windows
```

3. Install required dependencies.

```
pip install -r requirements.txt
```

6.3 Running the System

Run the following command to launch the full SafetyScan PPE Detection system:

```
Python main_system.py
```

Upon launching, the system's Graphical User Interface (GUI) will open in a pop-up window and display menu options:

- Launch Webcam: real-time detection using device camera.
- Upload Image File: support user upload for static testing.

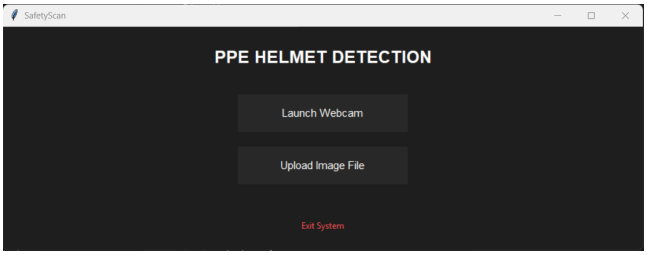

Figure 2. *SafetyScan Menu Interface*

The system should display the following expected outputs:

- YOLO: detected PPE objects (primarily helmets or heads).
- CNN: Helmet or No Helmet classification.
- RL: dynamic decision threshold.
- NLP: generated safety report.

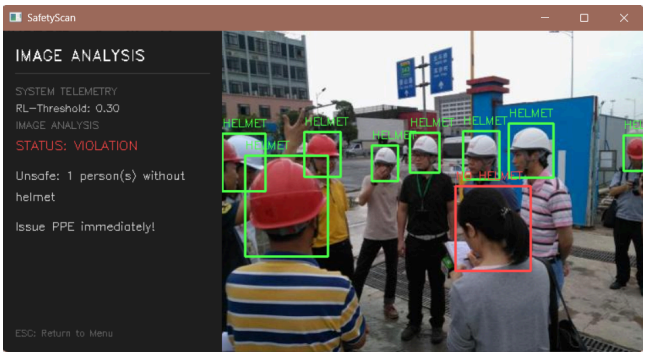

Figure 3. *SafetyScan Image Analysis Interface*

6.4 Pre-Trained Models

Pre-trained model files are readily available and downloadable from the provided GitHub repository.

- /models/: contains both YOLOv8 and CNN trained models.

6.5 Model Re-Training (Optional)

Model re-training is fully optional and is not required to run the system. If preferred, it can be reproduced using the Jupyter/Colab notebook provided in /notebooks/.

- YOLOv8 training with 50 epochs.
- Google Colab training with 15 epochs in ≤90 minutes completion time.

7. REFERENCES

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